

Visual Health Magazine Platform

Detailed, Outcome-Oriented Implementation Blueprint

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This blueprint converts the requested documentation set into detailed, measurable work packages. No responsible technology plan can guarantee human outcomes, medical results, or universal adoption. This plan therefore defines verifiable engineering, content, usability, safety, accessibility, and delivery outcomes, with explicit acceptance criteria and evidence for every milestone.

Executive operating model

The product

A multilingual visual health-learning publication that feels like a premium magazine while remaining usable as a simple, accessible, audio-assisted lesson system.

The delivery promise

Every release must prove five things:

1. **Understandable:** representative users can explain the lesson's main message.
2. **Actionable:** users can identify a safe next step or support pathway.
3. **Accessible:** the lesson works without 3D, colour-only cues, or audio.
4. **Trustworthy:** every medical claim is sourced and clinically reviewed.
5. **Reliable:** the core experience works on target low-end devices and weak networks.

Definition of outcome-oriented

A feature is not complete because code exists. It is complete only when:

- The intended user problem is stated.
- Acceptance criteria are testable.
- Automated checks pass.
- Manual verification is recorded.
- Representative users or reviewers validate it.

- Risks and limitations are documented.
- The project-memory file is updated.

1. Product Requirements Document

1.1 Strategic objective

Build a trusted, multilingual, low-literacy health education platform that improves understanding and encourages appropriate professional care without diagnosing users or collecting unnecessary sensitive information.

1.2 First-release outcome targets

These are targets to validate, not unverified promises:

Outcome	Target evidence	Release gate
Users understand the lesson	At least 80% of pilot participants explain the key message correctly	Independent comprehension test
Users identify a safe action	At least 80% select the appropriate next step in scenario testing	Facilitated usability test
Users can navigate independently	At least 90% complete language change and lesson restart without help	Task observation
Accessible fallback works	100% of core lesson information remains available in linear mode	Accessibility review
Core device performance	Core lesson remains usable on defined low-end Android test devices	Device test report
Clinical safety	100% of required medical claims have approved sources and reviewers	Publication checklist
Translation quality	No critical terminology or safety errors in launch languages	Native-language review

Do not report these as achieved until evidence is collected and stored.

1.3 Audience assumptions to validate

- Some users may read slowly or prefer audio.
- Some users may share devices.
- Some users may fear stigma or exposure.
- Many users may have intermittent connectivity.
- Local language may be preferred over English.
- Users may be unfamiliar with page-turn interfaces.
- Health workers may use the product in group settings.

1.4 Product requirements by priority

P0 — essential for a safe first release

- Anonymous access to published educational content.
- Linear reader.
- At least one reviewed publication.
- At least one reviewed language.
- Semantic text and visual alternatives.
- Audio transcript and replay.
- Clinical review workflow.
- Publication versioning and rollback.
- Security controls for staff access.
- Error and fallback states.

P1 — essential for a strong pilot

- CSS magazine reader.
- Touch navigation.
- Synchronized word and visual highlighting.
- Multiple languages.
- Local bookmarks and progress.
- Offline current-lesson support.
- Facilitator mode.
- Comprehension checks.

P2 — enhancement after validation

- Drag-to-turn physics.
- WebGL page curvature.
- Advanced particles and ambient effects.
- Optional accounts and synchronized dashboards.
- Expanded analytics.

1.5 User stories and acceptance criteria

Story: choose a language

As a reader, I want to select a language before reading so that I can understand the publication comfortably.

Acceptance:

- Language options have readable labels.
- The selected locale is reflected in the URL.
- The first lesson loads in that locale.
- Missing translation is handled clearly.
- Screen readers announce the change.
- The selection remains after reopening the application.

Evidence:

- Playwright test.
- Manual keyboard test.
- Native-language review record.

Story: understand through visuals

As a reader, I want an illustration to show the meaning of the explanation so that I do not depend on long text.

Acceptance:

- The visual has a short label and full description.
- Important targets can be focused.
- The narration focuses the relevant target.
- The same meaning exists in linear text.
- The visual remains understandable when animation is disabled.

Evidence:

- Accessibility test.
- Visual QA screenshot.
- Comprehension interview notes.

Story: listen and follow along

As a reader, I want audio to highlight the words and visual region being explained.

Acceptance:

- Play, pause, seek, speed, and replay work.

- Word timing has no invalid overlaps.
- Visual focus follows the current timing.
- Pausing audio pauses nonessential synchronized animation.
- Transcript remains available.
- Missing audio never removes text.

Evidence:

- Audio integration tests.
- Manual timing review.
- Browser recording from a representative device.

1.6 Product analytics with privacy

Use aggregated, non-sensitive events only where justified:

```
lesson_opened
lesson_completed
mode_selected
language_selected
audio_started
audio_completed
visual_fallback_used
technical_error
```

Do not record the precise sensitive topic, free-form text, diagnosis, or health information.
Analytics must not be necessary for the core learning journey.

1.7 Product risks

Risk	Impact	Probability	Mitigation	Owner
Visual effects distract from content	High	Medium	Test comprehension before polishing physics	Product
Automatic translation changes clinical meaning	High	High	Human native-language and clinical review	Content
Graphic imagery causes distress	High	Medium	Custom respectful illustrations and content review	Clinical
WebGL fails on low-end phones	Medium	High	Lite profile and DOM/SVG fallback	Engineering
Sensitive usage is exposed	High	Medium	Anonymous default and privacy review	Security
Content becomes outdated	High	Medium	Source register and re-review schedule	Clinical
Agent generates unsafe code or claims	High	Medium	Small tasks, tests, human approval gates	Engineering

2. Content and Clinical Governance

2.1 Governance objective

Create a repeatable content pipeline where clinical accuracy, local relevance, language quality, accessibility, and visual meaning are reviewed separately and recorded.

2.2 Claim-level workflow

For every factual statement:

1. Write the claim in plain language.
2. Identify the intended audience.
3. Attach an authoritative source.
4. Identify whether the claim is universal or location-specific.
5. Obtain clinical review.
6. Adapt the visual representation.
7. Review the visual for unintended meaning.
8. Translate only after the source wording is approved.
9. Review the translation with a native speaker.
10. Approve narration and timings.
11. Publish as a versioned release.

WHO provides current public-health information on STI scope, prevention, diagnosis, and treatment, but local guidance must be checked before publication. [web:27]

2.3 Required content fields

```
content_id
publication_id
chapter_id
lesson_id
block_id
source_claim
plain_language_claim
clinical_category
population_scope
country_scope
source_title
source_url
source_date
review_due_date
clinical_reviewer
clinical_status
translation_status_by_locale
narration_status_by_locale
accessibility_status
```

```
asset_license
publication_version
```

2.4 Medical image policy

Use a three-level visual classification:

Level A — abstract educational visual

Examples:

- Icons.
- Pathway diagrams.
- Human figures.
- Clinic journey.
- Colour-coded action cards.

Default choice for most lessons.

Level B — anatomical illustration

Allowed only when it improves understanding and has:

- Clinical justification.
- Respectful art direction.
- Age-appropriate presentation.
- Non-diagnostic wording.
- Accessibility description.
- Clinical approval.

Level C — clinical photograph

Use only when necessary for a clearly documented educational purpose and when licensing, consent, safeguarding, clinical review, and audience suitability are all resolved. Do not use external search images by default.

2.5 Content safety tests

Before approval, ask:

- Could the wording make a reader feel blamed or ashamed?
- Could a reader believe the visual proves a diagnosis?
- Could the reader misunderstand a prevention action as a guarantee?
- Does the advice vary by age, pregnancy, medication, or local policy?
- Is urgent-care wording needed?

- Is the referral information current?
- Does the translated phrase retain the same medical meaning?

2.6 Content release gate

A lesson can be published only when:

- All factual claims have sources.
- Clinical status is approved.
- Translation status is approved for every advertised locale.
- Narration matches approved text.
- Visual descriptions exist.
- SVG assets are sanitized.
- Accessibility review passes.
- Publication version is recorded.
- Re-review date is set.

2.7 Review service-level targets

Suggested operational targets:

- Critical safety issue: acknowledge within 4 working hours.
- Content correction: triage within 1 working day.
- Routine clinical review: complete within 5 working days after a complete submission.
- Translation review: complete within 5 working days after source approval.
- Scheduled content review: begin at least 30 days before expiry.

These are team service targets, not guarantees of medical review availability.

3. Technical Architecture Document

3.1 System boundaries

Inside the first product

- Published multilingual lessons.
- Reader modes.
- Visual scenes.
- Audio and transcripts.
- Content administration.
- Review workflow.

- Optional local progress.
- Privacy-preserving technical analytics.

Outside the first product

- Diagnostic algorithms.
- Individual clinical records.
- Prescriptions.
- Open medical chat.
- User-generated medical advice.
- Unverified external image collection.

3.2 Component architecture

Presentation

ReaderShell
MagazineReader
LinearReader
FacilitatorReader
SettingsPanel

Interaction

ReaderStateMachine
PageTurnController
AudioController
HighlightController
CapabilityController

Content

PublicationManifest
LessonRenderer
LocaleResolver
ContentValidator
ReviewStatusResolver

Visuals

SemanticSvgRenderer
StaticSceneFallback
WebglEnhancement
MotionPreferenceAdapter

Platform

ServiceWorker
LocalProgressStore
ErrorBoundary
PerformanceMonitor
PrivacyAnalytics

Backend

Content API
Review API

Publication API
Supabase/PostgreSQL
Storage

3.3 Suggested application routes

```
/[locale]
/[locale]/reader/[bookSlug]
/[locale]/reader/[bookSlug]/[chapterSlug]
/[locale]/learn/[lessonSlug]
/[locale]/help
/[locale]/privacy
/[locale]/accessibility
/admin/publications
/admin/lessons/[lessonId]
/admin/reviews/[reviewId]
/admin/releases
```

3.4 API contracts

Get publication manifest

```
GET /api/publications/{slug}?locale=hi&version=latest
```

Response:

```
{
  "publication": {
    "slug": "sexual-health-basics",
    "version": "2026.08.1",
    "locale": "hi",
    "status": "published"
  },
  "chapters": [],
  "assets": [],
  "expiresAt": null
}
```

Record optional progress

```
PUT /api/progress/{lessonId}
```

Requirements:

- Authenticated user only.
- Own-record policy.
- No sensitive free text.

- Idempotent update.
- User deletion support.

Review submission

```
POST /api/admin/reviews/{contentId}/decision
```

Requirements:

- Staff authentication.
- Role-specific authorization.
- Required decision reason.
- Audit event.
- No direct publication unless workflow permits it.

3.5 Page engine design

The page engine should expose a stable interface:

```
interface ReaderController {
  currentIndex: number
  totalPages: number
  mode: "magazine" | "linear" | "facilitator"
  next(): Promise<void>
  previous(): Promise<void>
  goTo(index: number): Promise<void>
  setMode(mode: ReaderController["mode"]): void
  setMotionPreference(value: "system" | "reduced" | "full"): void
}
```

The UI must not rely on querying arbitrary DOM elements to determine application state.

3.6 Synchronized media timeline

Use one timeline model for text, audio, and visual targets:

```
export type TimelineEvent = {
  id: string
  startMs: number
  endMs: number
  blockId: string
  textTokenIds?: string[]
  sceneTargetIds?: string[]
  action?: "highlight" | "pulse" | "trace" | "show"
}
```

Validation rules:

- `startMs >= 0`.
- `endMs > startMs`.
- Events belong to an existing block.
- Target IDs exist in the scene.
- Token IDs exist in the transcript.
- Critical event overlap is intentional and documented.

3.7 WebGL architecture

WebGL is optional

WebGL must be loaded lazily and destroyed when not needed. It should not block first content.

Scene contract

```
interface EnhancedScene {
  id: string
  preload(): Promise<void>
  mount(container: HTMLElement): void
  update(progress: number): void
  focus(targetId: string): void
  pause(): void
  dispose(): void
}
```

Fallback chain

```
WebGL enhanced scene
-> CSS/SVG balanced scene
-> static SVG/raster scene
-> semantic text explanation
```

3.8 Storage strategy

Separate buckets or prefixes:

```
public-publications/
public-assets/
draft-assets/
review-assets/
audio/{locale}/
exports/
```

Use immutable filenames containing the publication version or content hash.

3.9 Content-as-code versus CMS

Recommended first approach:

- Content schema in TypeScript.
- Content source in GitHub for traceability.
- Database publication metadata and review status.
- Admin interface only after the content model is stable.

This reduces early CMS complexity while preserving an upgrade path for non-technical editors.

3.10 Technology decision rules

Choose a library only if it improves one of:

- Comprehension.
- Accessibility.
- Performance.
- Reliability.
- Maintainability.
- Localization.
- Security.

Do not add WebGL, a physics library, or a CMS solely because it is fashionable.

4. Security and Access Document

4.1 Security posture

The application is public-facing and handles a sensitive subject. The safest default is to minimize user identity, minimize stored behaviour, and strictly separate public content from internal editorial data.

4.2 Roles and permissions matrix

Capability	Anonymous	Learner	Editor	Clinical reviewer	Language reviewer	Accessibility reviewer	Admin
Read published content	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Read drafts	No	No	Assigned	Assigned	Assigned	Assigned	Controlled
Create draft	No	No	Yes	Optional	No	No	Yes
Edit medical claim	No	No	Draft request	Review decision	Request clarification	No	Controlled

Capability	Anonymous	Learner	Editor	Clinical reviewer	Language reviewer	Accessibility reviewer	Admin
Approve clinical content	No	No	No	Yes	No	No	Controlled
Approve translation	No	No	No	No	Yes	No	Controlled
Approve accessibility	No	No	No	No	No	Yes	Controlled
Publish release	No	No	No	No	No	No	Yes/approved release role
View learner identity	No	Own only	No	No	No	No	Minimal and audited
Delete own account	N/A	Yes	Staff process	Staff process	Staff process	Staff process	Controlled

4.3 Authentication requirements

- Public reading requires no account.
- Staff accounts require MFA.
- Privileged actions require recent authentication.
- Sessions expire appropriately.
- Role changes are audited.
- Service accounts use short-lived credentials where practical.

4.4 Authorization requirements

Every protected mutation must verify:

1. Authenticated identity.
2. Active role.
3. Resource scope.
4. Workflow state.
5. Required review status.
6. Audit logging.

Never rely on client-side role checks.

4.5 Database rules

- All tables use least-privilege policies.
- Public queries return only published records.
- Draft and review records are protected.
- User records are owner-scoped.
- Admin queries are server-side and role-checked.
- Deletion and export requests are audited.

4.6 Threat-response controls

XSS through content

- Render plain text by default.
- Sanitize allowed rich text.
- Disallow arbitrary scripts and event attributes.
- Use a restrictive Content Security Policy.

Malicious SVG

- Parse and sanitize on upload.
- Remove scripts, external links, event handlers, and embedded unsafe content.
- Re-validate at publication time.
- Serve from a controlled asset origin.

Unauthorized draft access

- Public endpoints filter by publication status server-side.
- Add automated tests for anonymous draft requests.
- Use separate storage policies.

Account takeover

- MFA for staff.
- Rate limiting.
- Secure password or identity-provider policy.
- Session revocation.
- Suspicious-login alerts where supported.

Analytics privacy

- Avoid sensitive URLs and query strings.
- Do not send health-topic identifiers tied to users.
- Aggregate or sample technical telemetry.
- Document retention periods.

4.7 Security acceptance tests

- Anonymous user cannot access draft lesson JSON.
- Learner A cannot access Learner B's progress.
- Editor cannot approve a clinical review.
- Language reviewer cannot publish.
- Unsanitized SVG is rejected.
- Invalid publication cannot become public.
- Error response contains no secrets or internal paths.
- Revoked staff session cannot perform a protected action.

4.8 Privacy notice requirements

Explain in simple language:

- Reading can work without an account.
- What optional data is stored.
- Why technical analytics may be used.
- How to clear local data.
- How to request deletion.
- Who to contact about privacy.

4.9 Security incident targets

- P0 security incident acknowledged immediately by the on-call owner.
- Public exposure disabled as soon as safely possible.
- Credentials rotated when relevant.
- Incident record created within one working day.
- Corrective controls tested before closure.

5. Frontend Specification Document

5.1 Information hierarchy

Every screen must answer, in order:

1. Where am I?
2. What is the one important idea?
3. What does the visual show?
4. What can I do next?
5. How can I get help or learn more?

5.2 Screen specifications

Home screen

Must include:

- Clear title and purpose.
- Language selector.
- Start reading button.
- Listen option.
- Simple mode option.
- Privacy link.
- Help or support link.

Success criteria:

- A first-time user identifies how to start without assistance.
- Primary action remains visible on small screens.

Book cover

Must include:

- Publication title.
- Edition or update date where appropriate.
- Clear start action.
- Optional content-sensitivity notice.
- No overwhelming animation before interaction.

Lesson page

Must include:

- Chapter and lesson label.
- One primary message.
- Visual or explanatory block.
- Narration controls.
- Text alternative.
- Next action.
- Progress indication.

Help page

Must include:

- What the application can and cannot do.
- How to find professional care.
- Privacy guidance.
- Emergency disclaimer where appropriate.
- Language and accessibility controls.

5.3 Magazine page physics requirements

A page flip may include:

- Spine-origin rotation.
- Pointer-driven progress.
- Velocity-based commit.
- Snap-back cancellation.
- Dynamic shadow.
- Page thickness illusion.
- Subtle crease.
- Layered paper texture.

It must not:

- Trap focus.
- Hide the current reading content from assistive technology.
- Prevent scrolling.
- Require precise corner dragging.
- Fail without WebGL.

5.4 Focus management

After a page change:

- Announce the new chapter or page.
- Move focus to the page heading or first meaningful heading.
- Do not move focus unexpectedly during a simple visual highlight.
- Keep audio controls reachable.

5.5 Audio controls

- Use large, labelled controls.
- Show progress and duration.
- Provide a transcript button.
- Allow speed selection.
- Keep pause and replay nearby.
- Do not autoplay sensitive audio.

5.6 Error and fallback states

Missing translation

Show:

- A clear explanation.
- Available fallback language.
- Continue-reading action.
- Internal missing-translation event.

Missing audio

Show:

- “Audio is unavailable right now.”
- Transcript.
- Read-aloud browser option where appropriate.

Missing visual

Show:

- Text explanation.
- Scene label.
- Retry option.

WebGL unavailable

Silently use the balanced or lite visual treatment unless the user has selected enhanced mode and needs an explanation.

5.7 Performance budgets

Initial engineering budgets:

- Keep initial JavaScript focused on the first route.
- Lazy-load WebGL and heavy scenes.
- Avoid loading audio for unseen lessons.
- Use responsive media sizes.
- Avoid large uncompressed textures.
- Monitor memory after repeated page turns.

Final numeric budgets must be confirmed after device profiling.

6. Design System Document

6.1 Visual concept: “Editorial warmth with scientific clarity”

Blend:

- Magazine composition.
- Human illustration.
- Calm healthcare colours.
- Tactile paper cues.
- Precise diagram design.
- Simple action cards.

The design must feel premium without becoming elite or difficult.

6.2 Design tokens

```
:root {
  --font-body: "Inter", system-ui, sans-serif;
  --font-display: "Fraunces", Georgia, serif;
  --font-ui: "Inter", system-ui, sans-serif;

  --color-paper: #fffaf0;
  --color-paper-deep: #f2eadb;
  --color-ink: #182027;
  --color-ink-muted: #52616b;
  --color-teal: #0b6b67;
  --color-teal-dark: #064946;
  --color-orange: #f29f58;
```

```

--color-blue: #3b82a0;
--color-mint: #d8eee6;
--color-warning: #b86b27;
--color-danger: #a63d40;
--color-focus: #1d62d1;
--color-white: #ffffff;
--color-border: #d9dedb;

--space-1: 0.25rem;
--space-2: 0.5rem;
--space-3: 0.75rem;
--space-4: 1rem;
--space-6: 1.5rem;
--space-8: 2rem;
--space-12: 3rem;
--space-16: 4rem;

--radius-sm: 0.35rem;
--radius-md: 0.75rem;
--radius-lg: 1.5rem;
--shadow-page: 0 1rem 2.5rem rgb(24 32 39 / 0.18);
--shadow-card: 0 0.5rem 1.5rem rgb(24 32 39 / 0.1);
}

```

These are provisional tokens. Contrast, script support, and legibility must be validated before final release.

6.3 Typography requirements

- Use a body font with broad script coverage.
- Test actual glyph rendering in every supported language.
- Avoid thin weights for body text.
- Use high line height for low-literacy reading.
- Do not encode essential text inside images.
- Allow user text scaling without clipping.

6.4 Colour usage

- Teal: trust, guidance, primary navigation.
- Mint: calm explanation surfaces.
- Orange: attention and action.
- Blue: services and information.
- Warning: important caution, never blame.
- Danger: urgent notice only, never a diagnosis indicator.

Every colour-coded state must also use text, icon, shape, or pattern.

6.5 Motion language

- Page turn: tactile and deliberate.
- Visual reveal: gentle and explanatory.
- Target focus: soft pulse or glow.
- Action confirmation: brief and clear.
- Background: nearly static.

Support the reduced-motion preference. MDN documents prefers-reduced-motion as the mechanism for adapting animations to user accessibility settings. [web:23]

6.6 Illustration production pipeline

1. Sketch storyboard.
2. Review educational meaning.
3. Create vector scene.
4. Assign stable target IDs.
5. Create accessibility description.
6. Create static fallback.
7. Add animation timeline.
8. Review with clinical and language teams.
9. Optimize asset.
10. Test on target devices.

6.7 Component acceptance

A component is design-system complete when:

- It has tokens rather than arbitrary values.
- It has all interaction states.
- It has mobile behaviour.
- It has keyboard behaviour.
- It has accessible names.
- It has loading and error states.
- It has visual regression coverage where appropriate.
- It has usage guidance.

7. Feature Ticket and Delivery System

7.1 Ticket quality standard

Every ticket must contain:

- User problem.
- Desired outcome.
- Acceptance criteria.
- Technical scope.
- Content/clinical dependencies.
- Accessibility impact.
- Security impact.
- Test evidence.
- Files changed.
- Current blocker.
- Next action.

7.2 Definition of ready

A ticket is Ready only when:

- The user problem is clear.
- Acceptance criteria are testable.
- Dependencies are known.
- Design or content input is attached.
- Security and accessibility implications are considered.

7.3 Definition of done

A ticket is Done only when:

- Implementation is complete.
- Unit/integration tests pass.
- Browser verification passes where relevant.
- Accessibility review passes where relevant.
- Security review passes where relevant.
- Screenshots or recordings are attached for visual work.
- Documentation is updated.
- Project memory is updated.

- No unresolved P0/P1 defect remains.

7.4 Prioritized epic plan

EPIC FND — Foundation

- FND-001: Initialize Next.js and TypeScript project.
- FND-002: Configure validation, linting, formatting, and tests.
- FND-003: Implement content schemas.
- FND-004: Implement publication manifest.
- FND-005: Create project-memory update command or template.

EPIC CNT — Content

- CNT-001: Create first lesson outline.
- CNT-002: Add claim and source register.
- CNT-003: Add clinical review workflow.
- CNT-004: Add translation review workflow.
- CNT-005: Add publication release gate.

EPIC RDR — Reader

- RDR-001: Linear reader.
- RDR-002: Magazine reader.
- RDR-003: Page navigation.
- RDR-004: Touch gestures.
- RDR-005: Focus and announcements.
- RDR-006: Reader settings.

EPIC VIS — Visuals

- VIS-001: Semantic SVG renderer.
- VIS-002: Visual target focus.
- VIS-003: Static fallbacks.
- VIS-004: Timeline highlights.
- VIS-005: WebGL enhancement.

EPIC AUD — Audio

- AUD-001: Narration controller.
- AUD-002: Transcript support.
- AUD-003: Word-timing validator.
- AUD-004: Multi-locale audio.

EPIC L10N — Localization

- L10N-001: Locale routing.
- L10N-002: Message loading.
- L10N-003: Language switcher.
- L10N-004: Translation status.
- L10N-005: Font and layout QA.

EPIC SEC — Security

- SEC-001: Public content policies.
- SEC-002: Staff roles.
- SEC-003: SVG sanitization.
- SEC-004: Audit events.
- SEC-005: Security regression tests.

EPIC QA — Quality

- QA-001: Unit coverage.
- QA-002: E2E reader journey.
- QA-003: Accessibility tests.
- QA-004: Device performance matrix.
- QA-005: Clinical comprehension pilot.

7.5 Weekly operating rhythm

- Monday: prioritize tickets and confirm blockers.
- Daily: update active file, status, and next action.
- Midweek: run integration and browser checks.
- Review day: clinical, language, accessibility, and product review.
- End of week: update project memory, evidence links, risks, and next sprint.

8. Project Phases and Milestones

M0 — Documentation baseline

Deliverables

- All core documents.
- Approved problem statement.
- Audience and language assumptions.
- Clinical reviewers.
- Security model.
- Initial backlog.
- Risk register.

Measurable exit criteria

- Product owner signs PRD.
- Clinical owner signs governance workflow.
- Engineering owner accepts architecture.
- Security owner accepts access model.
- Three prototype lessons selected.

M1 — Comprehension prototype

Deliverables

- Three lessons.
- One language.
- Linear reader.
- Static SVG.
- Narration for at least one lesson.

Measurable exit criteria

- At least 80% of pilot participants explain each key message correctly.
- No critical clinical or safeguarding issue remains.
- Users can identify the action step.

M2 — Design system

Deliverables

- Tokens.
- Typography.
- Component library.
- Lesson templates.
- Accessibility patterns.

Measurable exit criteria

- Components pass keyboard and contrast review.
- Three lessons render without custom one-off styling.

M3 — Magazine reader

Deliverables

- CSS flipbook.
- Two-page desktop spread.
- Single-page mobile view.
- Keyboard and touch controls.
- Linear fallback.

Measurable exit criteria

- 100% of core content remains usable with motion disabled.
- No critical page-order issue.
- Page changes announce correctly.

M4 — Immersive learning

Deliverables

- Audio synchronization.
- Word highlighting.
- Visual target highlighting.
- Transcript.
- Replay and speed control.

Measurable exit criteria

- Manual timing review passes for every published narration.
- Automated timing validation passes.
- Users can identify the visual region being explained.

M5 — Multilingual pilot

Deliverables

- Approved launch locales.
- Human translations.
- Native narration.
- Locale-specific visual QA.

Measurable exit criteria

- No critical translation defects.
- Safety instructions have approved local wording.
- Text and audio remain synchronized per locale.

M6 — Offline and field readiness

Deliverables

- Current lesson caching.
- Offline indicator.
- Local progress.
- Low-end device profile.
- Facilitator mode.

Measurable exit criteria

- Pilot device can reopen the current lesson without network.
- User can understand whether content may be stale.
- Lite mode remains usable.

M7 — WebGL enhancement

Deliverables

- Optional realistic page physics.
- Enhanced/balanced/lite profiles.
- Capability detection.
- Performance report.

Measurable exit criteria

- WebGL failure does not break learning.
- Performance remains within approved budgets.
- Users do not perform worse in comprehension testing because of the effects.

M8 — Production launch

Deliverables

- Security sign-off.
- Clinical sign-off.
- Accessibility sign-off.
- Pilot findings resolved.
- Monitoring and rollback.
- Support runbook.

Measurable exit criteria

- No open P0/P1 defects.
- All release gates signed.
- Rollback tested.
- Project memory current.

9. Project Memory and Documentation Operations

9.1 Single source of operational truth

Maintain:

```
docs/project-memory.md
docs/release-log.md
docs/risk-register.md
docs/decisions/
docs/feature-tickets/
```

9.2 Required update after every agent task

Date:
Task ID:
Objective:
Files inspected:
Files created:
Files changed:
Tests run:
Browser checks:
Accessibility checks:
Security checks:
Performance checks:
Content/clinical impact:
Evidence:
Known limitations:
Blockers:
Decision required:
Next task:

9.3 Blocker protocol

A blocker must contain:

- Exact description.
- Affected milestone.
- Affected files.
- Why work cannot continue safely.
- Owner needed to resolve it.
- Proposed options.
- Deadline or review date.

Do not hide blockers by marking a ticket Done with a note.

9.4 Decision protocol

For every material decision:

1. Describe the context.
2. List alternatives.
3. Explain the chosen option.
4. Explain consequences.
5. Record the decision date.
6. Identify who approved it.

9.5 Evidence storage

Attach or link:

- Test reports.
- Browser recordings.
- Accessibility findings.
- Device profiles.
- Clinical approval records.
- Translation review records.
- Security test results.
- Release manifests.

10. Testing and Quality Plan

10.1 Quality strategy

Test from four perspectives:

1. Can the user understand it?
2. Can the user operate it?
3. Is the content safe and accurate?
4. Does it work reliably on the target device and network?

10.2 Test matrix

Area	Automated	Manual	Representative-user validation
Content schema	Yes	Yes	No
Clinical wording	No	Clinical review	Yes where possible
Translation	Partial	Native review	Yes
Keyboard access	Yes	Yes	Optional
Screen reader	Partial	Yes	Yes
Page physics	Partial	Yes	Yes
Audio timing	Yes	Yes	Yes
Device performance	Partial	Yes	Yes
Privacy	Yes	Yes	No
Comprehension	No	Facilitated	Yes

10.3 Comprehension protocol

Use a moderated protocol with:

- At least two audience segments.
- Multiple language groups where possible.
- Neutral prompts.
- No correction during first response.
- Observation of hesitation and misinterpretation.
- Immediate feedback capture.
- Repeat testing after revision.

10.4 Performance protocol

Test:

- Cold load.
- Warm load.
- First lesson.
- Long chapter.
- Repeated page turns.
- Audio start under slow network.
- WebGL enabled and disabled.
- Memory after 10, 25, and 50 page transitions.

10.5 Release scorecard

Use a scorecard with status:

```
Clinical accuracy: Pass / Fail / Pending
Translation quality: Pass / Fail / Pending
Accessibility: Pass / Fail / Pending
Security: Pass / Fail / Pending
Performance: Pass / Fail / Pending
Offline: Pass / Fail / Pending
Content licensing: Pass / Fail / Pending
Comprehension: Pass / Fail / Pending
Rollback: Pass / Fail / Pending
```

No release is “green” while a required category is Pending.

11. Accessibility Plan

11.1 Standard

Target WCAG 2.2 AA as the engineering baseline. WCAG requires text alternatives for non-text content and supports transforming content into speech, symbols, or simpler language, which is particularly important for this project. [web:30]

11.2 Accessibility acceptance criteria

- All core functions work with keyboard only.
- Focus remains visible.
- Page changes are announced.
- All meaningful images have alternatives.
- SVG targets have accessible names.
- Audio has captions or transcripts.
- Reduced motion removes or substantially reduces nonessential animation.
- Text can be enlarged without loss of content.
- Colour is not the only indicator.
- Linear mode works without 3D.

11.3 Assistive technology test set

Validate with:

- Keyboard-only navigation.
- TalkBack on Android.
- VoiceOver on iOS.
- NVDA or equivalent desktop screen reader.
- Browser zoom and text scaling.
- High-contrast or forced-colour settings.

11.4 Motion control

Provide:

- System setting.
- Reduced motion.
- Full motion.
- Lite visual profile.
- Pause for background animation.

Use `prefers-reduced-motion` as a baseline signal and provide an in-app override. [web:23]
[web:24]

12. Localization Plan

12.1 Localization objective

Make every part of the experience—not just visible paragraphs—localizable:

- Navigation.
- Visual labels.
- Audio.
- Captions.
- Quizzes.
- Error states.
- Privacy notices.
- Referral instructions.
- Metadata.
- Accessibility descriptions.

12.2 Language rollout

Do not choose languages based only on technical convenience. Select them using:

- Audience need.
- Program geography.
- Health-worker recommendation.
- Available reviewers.
- Audio resources.
- Font quality.
- Referral-service availability.

12.3 Locale quality gates

A locale is launch-ready only when:

- All P0 content is translated.
- Clinical terminology is approved.
- Audio exists for the intended lessons.
- Captions match audio.
- Visual labels fit.

- No critical text falls back silently.
- Native reviewer signs off.

12.4 Internationalization implementation

Use locale-aware routing and message loading. `next-intl` supports localized rendering, formatting, and locale-based routing in Next.js applications. [web:34][web:37]

Recommended structure:

```
messages/  
  en.json  
  hi.json  
  bn.json  
content/  
  en/  
  hi/  
  bn/  
audio/  
  en/  
  hi/  
  bn/
```

12.5 Translation QA automation

Automate checks for:

- Missing keys.
- Duplicate keys.
- Placeholder mismatch.
- Unsafe HTML.
- Excessive text length.
- Missing visual target labels.
- Missing narration timing.
- Locale code mismatch.

13. Content Authoring Guide

13.1 Editorial formula

Every lesson should follow:

```
Question or concern  
-> Simple explanation  
-> Visual demonstration  
-> Myth correction
```

```
-> Practical action  
-> Help pathway
```

13.2 One-screen rule

Each screen should have one dominant idea. If a page requires several paragraphs of explanation, split it into multiple visual steps.

13.3 Lesson authoring template

```
id: lesson-example  
slug: example  
objective: "The reader can explain the main idea in their own words."  
key_message: "One approved plain-language message."  
blocks:  
  - id: opening  
    type: text  
  - id: visual-explanation  
    type: illustration  
    scene_id: scene-example  
  - id: action  
    type: action  
sources:  
  - source_id: source-example  
reviews:  
  clinical: pending  
  language: pending  
  accessibility: pending
```

13.4 Visual narration pattern

Use:

1. "Look at this picture."
2. "This part shows ..."
3. "Notice that ..."
4. "This means ..."
5. "A safe next step is ..."

13.5 Content examples for the first prototype

Lesson A — What is an STI?

Visual: abstract person-to-person connection map.

Outcome: reader explains that infections can pass between people through specific types of contact and that health workers can provide guidance.

Lesson B — Not every infection is visible

Visual: friendly figures with an optional hidden information layer.

Outcome: reader understands that appearance alone is not a reliable way to know and that testing or professional advice may be needed.

Lesson C — What happens during testing?

Visual: calm clinic journey.

Outcome: reader can describe the basic steps and identify where to seek help.

All final wording must be clinically reviewed.

14. Deployment Runbook

14.1 Branching

Recommended branches:

```
main          production-ready
staging        release candidate
feature/*      individual work
content/*      content changes
hotfix/*       urgent correction
```

14.2 Pull request requirements

Every pull request must include:

- Summary.
- Related ticket.
- Screenshots or recording for UI changes.
- Tests run.
- Accessibility impact.
- Security impact.
- Performance impact.
- Content or clinical impact.
- Rollback considerations.

14.3 CI pipeline

```
Install
-> Type check
-> Lint
-> Unit tests
-> Content validation
-> Security checks
-> Build
-> E2E smoke tests
-> Accessibility smoke tests
-> Preview deployment
```

14.4 Staging verification

Verify:

- Public and staff routes.
- All launch locales.
- Reader modes.
- Audio.
- SVG scenes.
- WebGL fallback.
- Offline current lesson.
- Error states.
- Analytics failure isolation.

14.5 Production release

1. Confirm release scorecard.
2. Confirm publication version.
3. Confirm rollback target.
4. Deploy application.
5. Publish content manifest.
6. Run smoke test.
7. Monitor technical metrics.
8. Record release.
9. Notify reviewers and support owners.

14.6 Rollback

Rollback must be possible for:

- Application version.
- Publication version.
- Individual lesson.
- Individual translation.
- Individual asset.
- Audio track.

14.7 Monitoring

Monitor:

- Error rate.
- Time to interactive.
- Asset failure rate.
- Audio failure rate.
- Reader fallback rate.
- WebGL failure rate.
- Offline cache errors.
- Unauthorized access attempts.
- Translation-missing errors.

Do not monitor sensitive reading topics at user identity level.

15. Antigravity Implementation Protocol

Google Antigravity is an agentic development platform and should be used as a controlled execution partner, not as an autonomous publisher. [web:15][web:21][web:26]

15.1 Agent rules

- Read the relevant documentation before coding.
- Work on one ticket or tightly related slice at a time.
- Do not invent medical claims.
- Do not publish content.
- Do not weaken security or accessibility to make tests pass.
- Run tests and browser verification.
- Record files and blockers.
- Ask for approval when a decision changes architecture or safety.

15.2 First Antigravity prompt

Act as a principal software architect, product manager, accessibility specialist, security engineer, health-content systems designer, and senior Next.js engineer.

First inspect the repository and read every file under docs/. Do not write production application code yet.

Tasks:

1. Compare the documentation for contradictions and missing decisions.
2. Produce docs/documentation-audit.md.
3. Produce docs/risk-register.md.
4. Produce docs/implementation-dependency-map.md.
5. Update docs/09-project-memory.md with findings.
6. Identify decisions that require human approval.
7. Do not invent clinical content.
8. Do not change the privacy model without approval.
9. Do not introduce WebGL into the first comprehension prototype.

Return an artifact containing files changed, unresolved questions, blockers, and the recommended next ticket.

15.3 Foundation prompt

Implement FND-001 through FND-005 from the approved documentation.

Requirements:

- Next.js App Router and TypeScript.
- Strict type checking.
- Zod content schemas.
- Content validation CLI.
- Unit tests.
- Environment validation.
- No production medical claims; use clearly marked placeholders.
- Update feature ticket status and project memory.
- Run type checks, lint, tests, and build.
- Report exact evidence and unresolved limitations.

15.4 Reader prompt

Implement RDR-001 and RDR-005.

Build the linear reader first. It must contain semantic HTML, keyboard navigation, focus management, accessible visual descriptions, transcript support, and reduced-motion behaviour. Do not make WebGL or page physics a dependency.

Then add a controlled CSS magazine mode while preserving linear mode as a complete fallback. Test mobile, keyboard, reduced motion, and missing-asset states.

15.5 Visual prompt

Implement VIS-001 through VIS-004.

Use semantic SVG scenes with stable target IDs, accessible labels, static fallbacks, and synchronized highlight events. Essential educational meaning must remain available in DOM or text. Validate all target and timing references before rendering.

15.6 WebGL prompt

Implement VIS-005 as a progressive enhancement experiment only.

Create one realistic page-physics scene. Add capability detection and enhanced, balanced, and lite modes. Do not block first content. Do not put essential text or health instructions only in WebGL. Measure performance on target devices and update the decision record with evidence.

15.7 Agent completion report

Every task must end with:

Task:
Status:
Files changed:
Tests run:
Browser verified:
Accessibility verified:
Security verified:
Performance checked:
Content/clinical impact:
Known limitations:
Blockers:
Recommended next task:

16. Launch Readiness Scorecard

Product

- ☐ Core audience validated.
- ☐ Key lessons understood by pilot users.
- ☐ Safe next action is clear.
- ☐ Anonymous reading works.

Clinical and content

- ☐ All claims sourced.
- ☐ Clinical review complete.
- ☐ Referral information current.
- ☐ Image and asset rights documented.
- ☐ Re-review dates set.

Accessibility

- ☐ Linear mode complete.
- ☐ Keyboard journey passes.
- ☐ Screen-reader journey reviewed.
- ☐ Captions/transcripts complete.
- ☐ Reduced motion works.
- ☐ Contrast and scaling pass.

Localization

- ☐ Launch languages approved.
- ☐ Translation reviewed.
- ☐ Narration reviewed.
- ☐ Visual labels reviewed.
- ☐ No critical fallback defects.

Security

- ☐ RLS tested.
- ☐ Staff MFA enabled.
- ☐ SVG sanitization tested.
- ☐ Secrets scan clean.
- ☐ Privacy notice approved.
- ☐ Incident process documented.

Engineering

- ☐ Type checks pass.
- ☐ Tests pass.
- ☐ Build passes.
- ☐ Target device profile passes.
- ☐ Offline current lesson passes.

- ☐ Rollback tested.
- ☐ Monitoring active.

Documentation

- ☐ Project memory current.
- ☐ Feature tickets current.
- ☐ Release log complete.
- ☐ Architecture decisions recorded.
- ☐ Runbook reviewed.

17. Immediate next actions

1. Confirm launch geography and first languages.
2. Name the clinical reviewer and language reviewers.
3. Approve the first three prototype lessons.
4. Create the repository from this documentation baseline.
5. Run the Antigravity documentation audit prompt.
6. Resolve blockers before production coding.
7. Implement the linear comprehension prototype.
8. Test with intended users.
9. Only then implement advanced magazine physics and WebGL.